MIL-E-1/1021C
3 October 1977
SUPERSEDING
MIL-E-1/1021B
7 July 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5Y3WGTA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, full-wave, high vacuum

Outline --- 9-13 (EIA)

Base --- B5-10 or B5-62 (phenolic)

Envelope --- T-9

Cathode --- Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	nc	f	-	2a	-	1a	-	f

ABSOLUTE RATINGS:

Parameter:	Ef	Epp/a	epx	Rl	Cl	Io	ib/a
Unit:	V	Vac	v	Ohms	µf	mAdc	ma
Maximum:	5.5	---	1,550	--	--	(See note 1)	415
Minimum:	4.5	---	---	--	--	----	---
TEST CONDITIONS:	5.0	400	---	2,750	4	----	---

ABSOLUTE RATINGS:

Parameter:	i surge/a	TE	Alt
Unit:	a	°C	ft
Maximum:	2.0	225	65,000
Minimum:	---	---	(See note 2 and figure 1)
TEST CONDITIONS:	---	---	---

GENERAL:

Qualification - Required

Reliable tube

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1201	Short and Discontinuity detection		0.4	II	---	---	---	---
1231	Emission current	Eb = 75 V See note 3	0.65	II	Is	120	---	mAdc
1301	Filament current		0.65	II	If	1.6	2.0	A
1353	Operation of rectifiers (1)	See note 4	----	--	Io	---	---	mAdc
1353	Operation of rectifiers (1)	See notes 4 and 5	0.65	II	Io	130	150	mAdc
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	No voltages applied	6.5	Code H	---	---	---	---
1353	Operation of rectifiers (2)	Ef = 4.5 V See note 4	2.5	S3	ΔIs Ef	---	10	%
1211	Insulation of electrodes		2.5	S3	---	---	---	---
1101	Secureness of base, cap, or insert		2.5	S3	---	---	---	---
1111	Base pin solder depth		---	----	---	---	---	---
1041	Shock	450G; Epp/a = 0	---	---	---	---	---	---
1031	Vibration fatigue	fixed frequency F = 25 min, 60 max	6.5	See note 6	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (Percent Defective)	Insp Level or Code	Allowable Defectives Per Characteristic		Sym	LIMITS		Unit
					1st sample	Combined samples		Min	Max	
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 5.5 V; Eb = 0; 1 min "on", 4 min "off"	---	---	---	------	---	---	---	---
1521	Survival-rate life	TA = room See note 7	---	---	---	------	---	---	---	---
----	Survival-rate life test end points:									
1201	Short and discontinuity detection		---	---	---	------	---	---	---	---
1353	Operation of rectifiers (1)		---	---	---	------	Io	115	---	mAdc
1501	Intermittent life	TE = 225°C (min) See note 8	---	---	---	------	---	---	---	---
----	Intermittent life test end points: (500 hours)									
----	Inoperatives		---	---	1	3	---	---	---	---
1301	Filament current		---	---	1	3	If	1.5	2.1	A
1353	Operation of rectifiers (1)		---	---	1	3	Io	115	---	mAdc
1353	Change in operation of rectifiers (1) of individual tubes		---	---	1	3	ΔIo t	---	10	%
1211	Insulation of electrodes		---	---	2	5	R	50	---	Meg
----	Total defectives		---	---	4	8	---	---	---	---
----	Intermittent life test end points: (1,000 hours)									
----	Inoperatives		---	---	2	5	---	---	---	---
1301	Filament current		---	---	2	5	If	1.45	2.15	A
1353	Operation of rectifiers (1)		---	---	2	5	Io	110	---	mAdc
----	Total defectives		---	---	5	10	---	---	---	---

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NOTES:

1. To simplify the application of the maximum ratings to circuit design, the electrical maximum ratings are also in chart form as rating charts I, II, and III. Rating chart I presents the maximum ratings for ac anode-supply voltage and dc output current. Rating chart II provides a convenient method for checking conformance with the maximum steady-state peak-anode-current rating. Rating chart III offers a convenient method for checking conformance with the maximum peak-anode-current rating.

   With a capacitor-input filter, the conditions of rating charts I, II, and III shall be satisfied to obtain performance within all of the electrical maximum ratings. With a choke-input filter, operation within the indicated boundary of rating chart I will assure performance within all of the appropriate electrical maximum ratings.

   The ratings as presented in chart form are on the basis of the system of absolute-maximum ratings. In addition, the values of ac supply voltage as presented refer to the unloaded supply voltages.

   Rating chart I (see figure 1). To determine whether a given set of operating conditions is within the maximum ratings presented in rating chart 1, the operating point of ac supply voltage (rms) per anode and dc output current per tube for the application shall be entered on the chart. If, for choke-input-filter operation, this point falls within the boundary FABCDG, operation within ratings of ac supply voltage and dc output current is assured. For capacitor-input-filter operation, the operating point shall fall within the boundary FAEDG. If the operating point falls outside of the prescribed area of permissible operation either the dc output current or the ac supply voltage shall be reduced until operation within the area of permissible operation is obtained. The type of corrective action to be taken necessarily depends on the requirements of the application.

   Rating chart II (see figure 2). To determine whether a given set of operating conditions is within the maximum ratings presented in rating chart II, the operating point of dc output current per tube and rectification efficiency shall be entered on the chart. If this point falls within the shaded area of the chart, operation within the maximum steady-state peak-anode-current is assured. If the operating point falls outside of the prescribed area of permissible operation, corrective action shall be taken. For a given ac supply voltage and dc output current, the rectification efficiency can usually be reduced either by increasing the total supply resistance per anode or by decreasing the filter input capacitor.

   Rating chart III (see figure 3). To determine whether a given set of operating conditions is within the maximum ratings presented in rating chart III, the total Rs is first calculated from measurements of Rpri, Rsec, RA, and N. The operating point of Rs and ac supply voltage is then entered on rating chart III. If the total anode-supply resistance is equal to or greater than the minimum value of Rs, as indicated by the curve, operation with the maximum transient peak-anode-current rating is assured. If the conditions of rating chart III are not satisfied, sufficient additional resistance shall be added in series with each diode anode so that the indicated minimum value of Rs is obtained.

   It should be noted that the anode-supply resistance serves to limit the steady-state peak current and the transient peak current. The minimum value of resistance required for these two functions will not necessarily be the same. The larger value of resistance shall be used.

   Series inductance in the anode supply is also effective in limiting the transient peak-anode current. If appreciable series inductance is present in the anode supply, a smaller-than-indicated value of series-anode-supply resistance may be used, providing it is determined experimentally that the combined effect of the available supply resistance and inductance is adequate to limit the transient peak-anode current to its rated maximum value.

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NOTES: -Continued

2. The maximum voltage appearing between any pair of pins shall not be greater than the peak anode voltage rating. If altitude rating is exceeded, reduction of instantaneous voltages (Ef excluded) may be required.

3. Test each unit separately.

4. In a full-wave circuit, adjust Zp/a so that a bogey tube gives Io = 140 mAdc and ib not less than 375 ma per anode. A bogey tube is defined as a tube with a voltage drop of Etd = 60 Vdc at Is = 125 mAdc per anode.

5. This test to be performed at the conclusion of the holding period.

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot passes, the 6-month rule shall apply. MIL-STD-105, sample code letter E, shall apply.

7. In a full-wave life-test circuit, the values specified for Rl and CL may be considered as approximate and shall be adjusted initially to give not less than Io = 140 mAdc and ib = 375 ma per anode with a bogey tube.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (see method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

Custodian:

Army - EL
Navy - EC
Air Force - 85

Review activities:

Army - MU, MI
Air Force - 99
DSA - ES

User activities:

Army - ME, WC
Navy - AS, OS, MC, CG,
Air Force - 11

Preparing activity:

Air Force - 85

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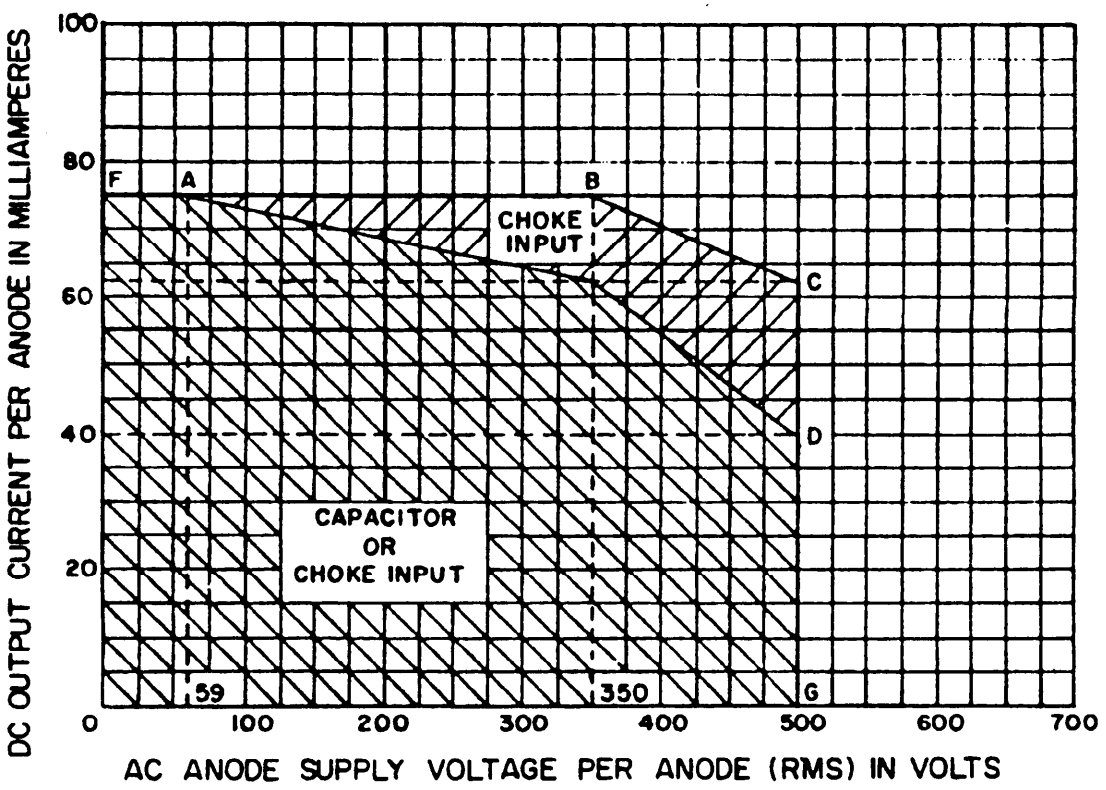


FIGURE 1. Rating chart I.

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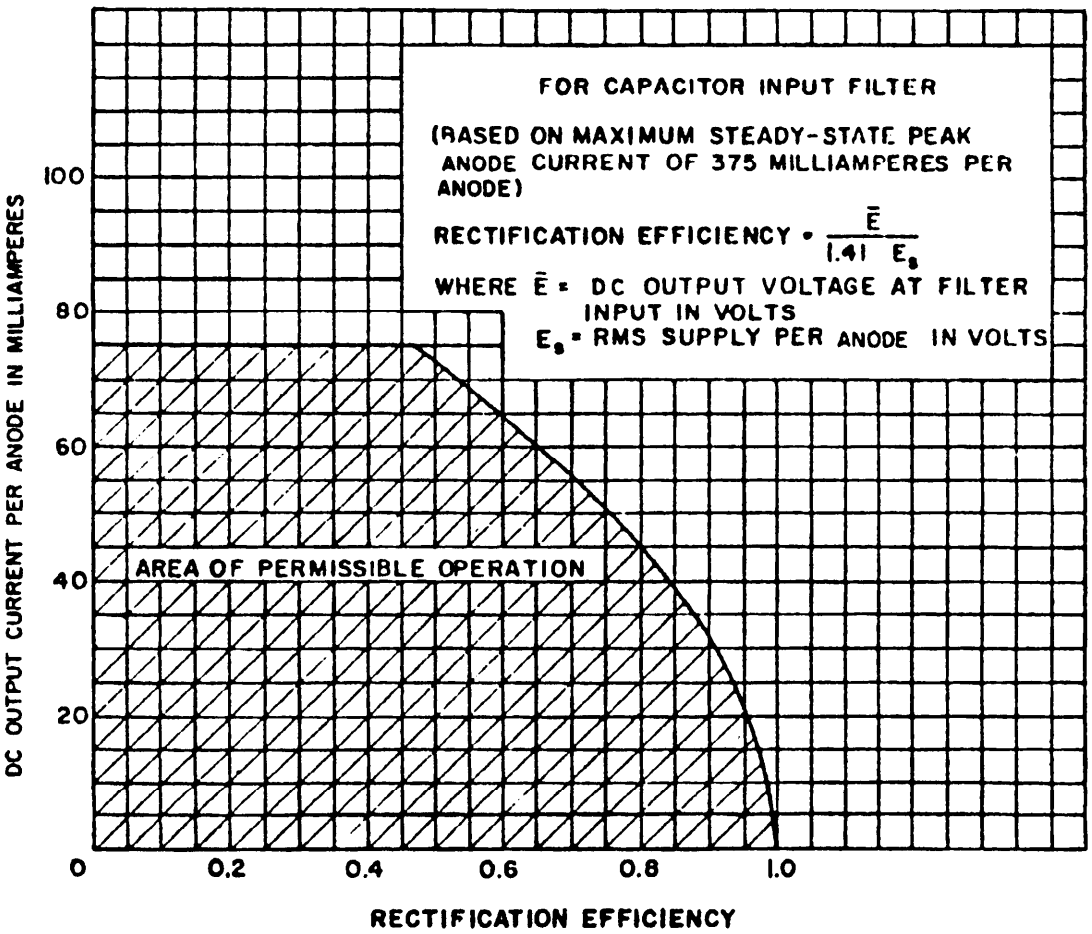


FIGURE 2. Rating chart II.

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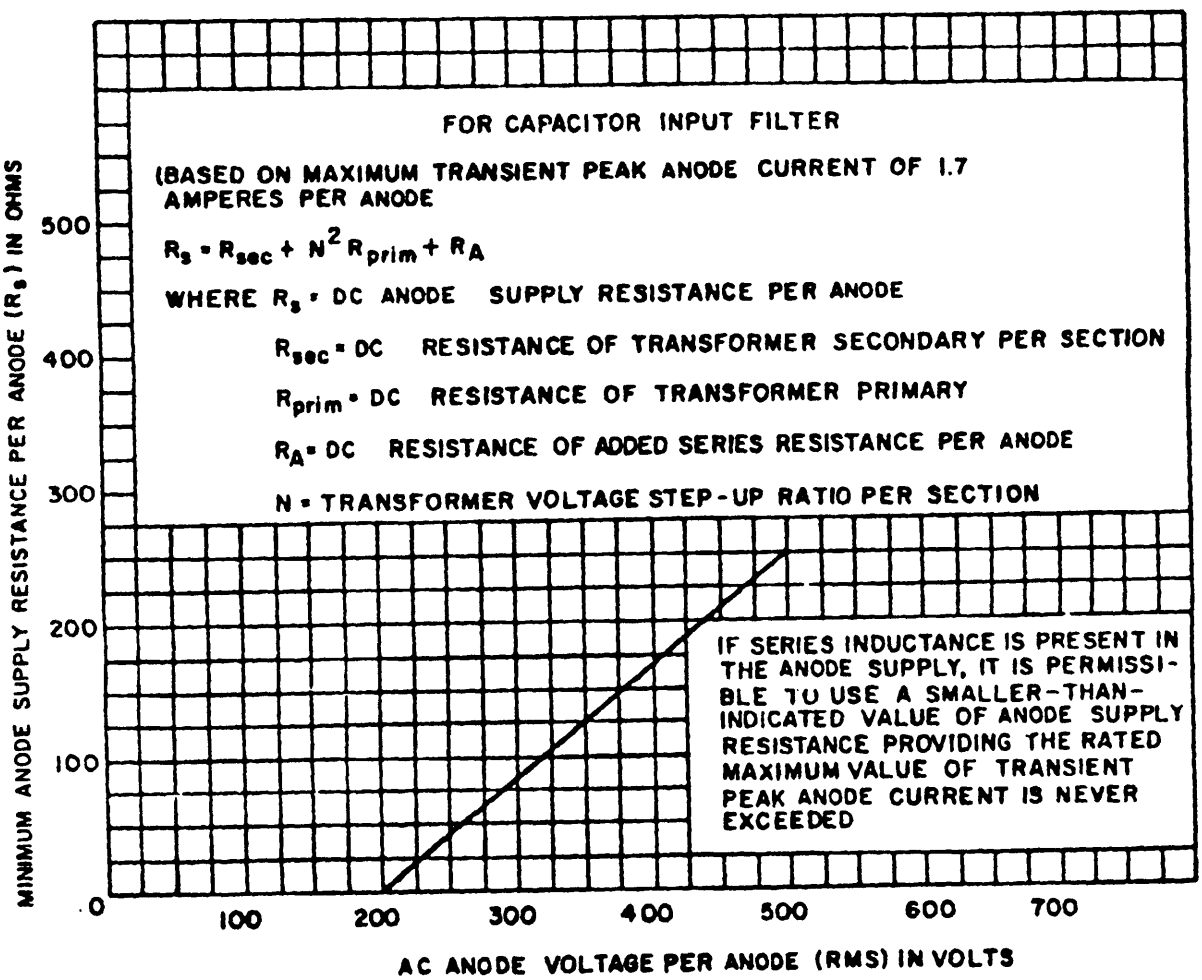


FIGURE 3. Rating chart III.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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